SDG 9: INFRASTRUCTURE

Live Pulse

Project Focus: Addressing SDG 9 (Industry, Innovation and Infrastructure) by focusing on public transport reliability and digital service monitoring.

The Core Problem: Organizations usually notice infrastructure problems only after they hit the news. Passengers write about broken lifts, delayed trains, and crowding every day, but no human team can read all of it manually.

Target User & Action: "Maaike," an Operations Manager at NS (Nederlandse Spoorwegen). She monitors the real-time "pulse" to identify where crowding or delays hit hardest before an official report is filed, allowing her to dispatch support staff proactively.



The Solution Architecture & Data Pipeline

Continuous Pipeline: A Python script fetches new items every hour, analyzes them, and stores them with a timestamp, running completely autonomously without manual button-pressing.

Live Data Stream: Pulling real-time, public text from Mastodon hashtag timelines and Reddit RSS feeds (specifically tracking #ns, #storing, and #trein).

Privacy & Ethical Data Handling: The system strictly stores only what is needed: the text, the time, and the sentiment score. All usernames, profile data, and personally identifiable information are immediately stripped so nobody can be singled out.

Automation Infrastructure: Orchestrated entirely via GitHub Actions to maintain a reliable, continuous background feed.

AI Sentiment Engine & The "Honest Test"

AI Engine Selection: Powered by RobBERT, a pre-trained Dutch sentiment transformer loaded from Hugging Face and run entirely by our own code, explicitly avoiding black-box chatbot APIs.

The "Honest Test" Methodology: We manually labeled 52 real posts directly from our data stream to establish a reliable ground truth for system evaluation.

Model Comparison: RobBERT was rigorously tested alongside Cardiff RoBERTa (an English model) and a simple word-list baseline. RobBERT successfully captured linguistic nuances, sarcasm, and Dutch transit jargon that the baselines missed.

Ethical Blindspots: We acknowledge that the stream only captures vocal digital users; older demographics or those who do not post online are missing from this sentiment pulse.

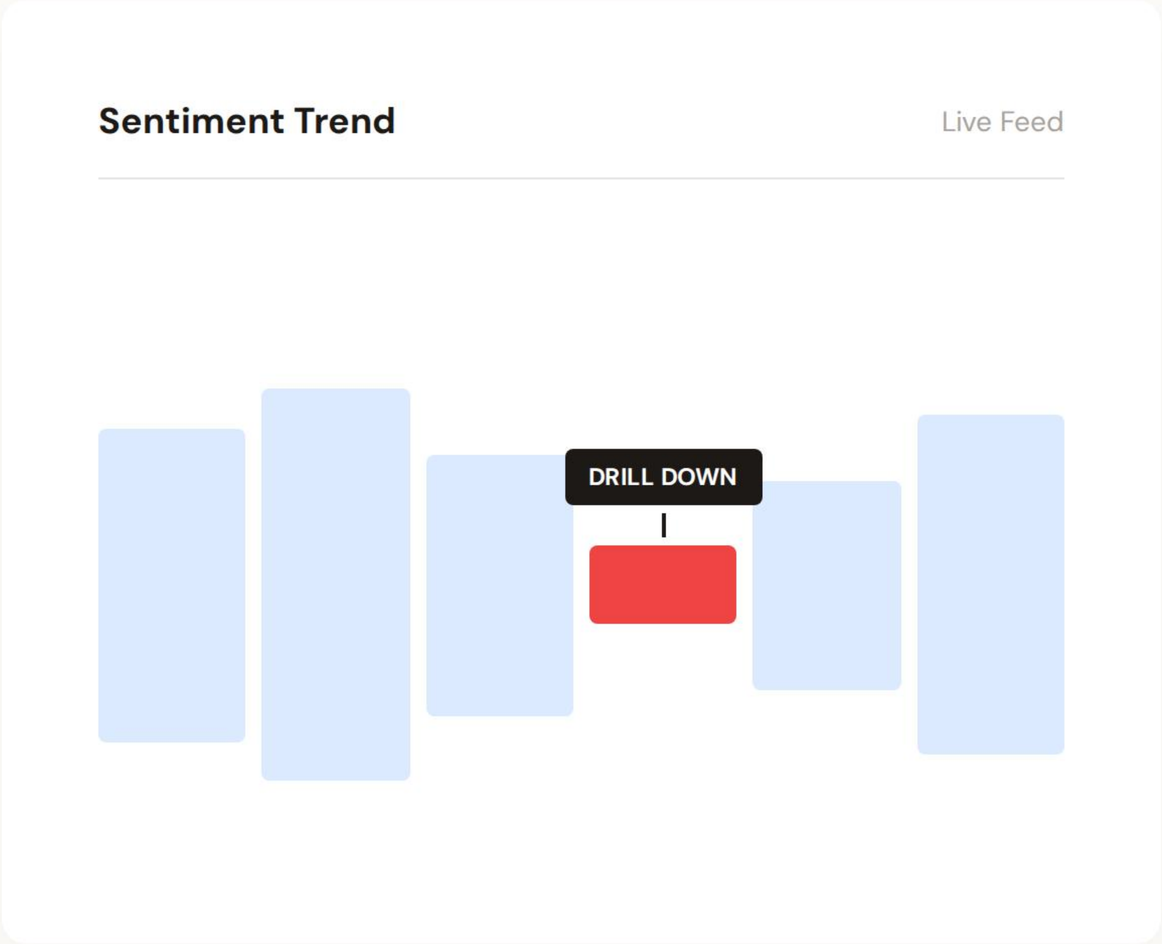
Data Storage & Interactive Dashboard

Data Storage & Management: All processed text and sentiment scores are logged in a local SQLite database (pulse.db) and continuously exported as JSON.

Accessible Visualization: A web-based dashboard built with app.py and index.html displays the sentiment per hour in a clear chart, designed specifically so someone who is not a programmer can read it.

Root-Cause Drill-Down: When the dashboard shows a sudden wave of negative sentiment, the user can click directly on the dip to reveal the exact items behind the peak.

Problem-Solution Fit: By connecting the automated sentiment score to the underlying posts, Maaike can verify the exact cause of frustration (e.g., a broken ticket machine) and make a concrete operational decision.



From reactive response to proactive operation.

Empowering dispatch with real-time commuter truth.

Image Sources



https://delta.tudelft.nl/wp-content/uploads/2023/12/VolleTrein_Conducteur_Nederlandse_Spoorwegen_WikimediaCommons_1600px.jpg

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